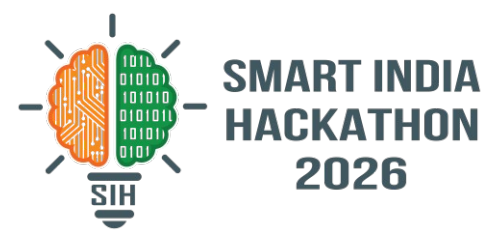


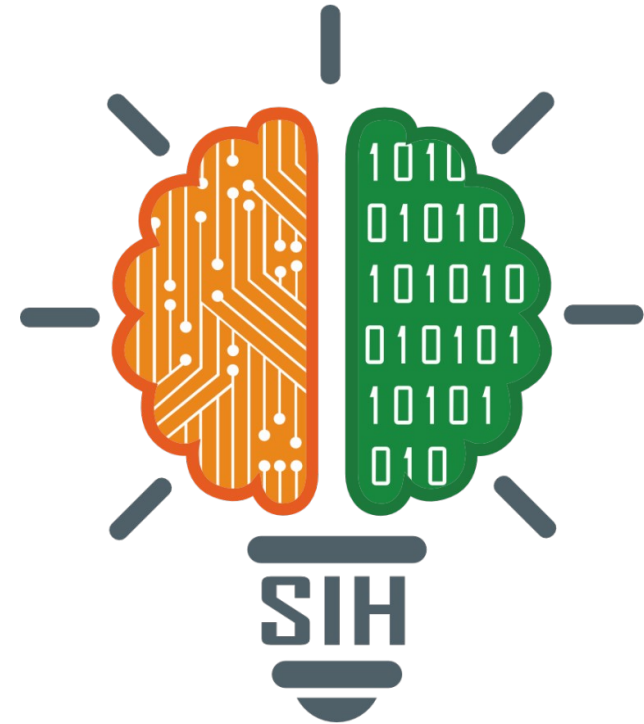


# SMART INDIA HACKATHON 2026



## TITLE PAGE

- **Problem Statement ID** – SIH26153
- **Problem Statement Title** – AI based Network Attack Forecasting from Network Traffic Data
- **Theme** – Blockchain and Cybersecurity
- **PS Category** – Software
- **Team ID** – 149775
- **Team Name (Registered on portal)** – Resurrección





# NetSentinel Foresight : A World Model that Forecasts Network Attacks



Enterprise & Critical Information Infrastructure | Learns  $P(S_{t+1} | S_t)$  | K-step attack forecasting | MITRE ATT&CK stages | SHAP + attention explanations | Runs fully offline

## IDEA / SOLUTION

Intrusion detectors tell you a breach has happened. **NetSentinel Foresight tells you where the attacker is going next, while the attacker is still scanning and there is still time to stop them.**



### Learns how the network changes over time

A world model learns  $P(S_{t+1} | S_t)$ : how a graph of hosts, flow and packet features changes each minute. It does not just label single flows.



### Plays the future forward

Simulates 64 possible futures up to K windows ahead. The share that reach an infiltration state is the infiltration probability, shown with a confidence band.



### Names the next attack stage and target

Each forecast maps to MITRE ATT&CK: Reconnaissance → Initial Access → Lateral Movement → C2 → Exfiltration, and names the host likely hit next.



### Shows its reasoning

Attention weights and SHAP values list the flags, ports and flow statistics behind every forecast. No unexplained scores.



### “What if we block it?”

The defender applies an action (block port 445, isolate a host) inside the model and sees how the risk changes, before touching the real network.



### A working prototype, not only a plan

PCAP or CSV in; forecasts, Shapley explanations, flagged flows and signed receipts out. Offline app, ordinary CPU.

## PROBLEM RESOLUTION

- **Classifiers see flows, not campaigns:** a benign/malicious label per flow throws away order and timing. A slow scan, then a brute-force, then lateral SMB traffic is one story, not three unrelated alerts.
- **Alerts come after the damage:** detection fires once a stage has already happened. Defenders need the warning while the attacker is still in reconnaissance.
- **Scores nobody can act on:** a probability with no “why” and no “what next” gets ignored by SOC analysts and cannot be audited in CII.

## INNOVATION & UNIQUENESS



### Attack “weather forecast”

Many sampled futures give a risk band and time-to-compromise, not one label.



### Flow + packet, on a graph

TTL, window size and port order catch slow scans; the graph predicts which host is next.



### Auditable forecasts

Each forecast is signed into a Merkle ledger and later scored against what actually happened.



### Knows the attacker’s playbook

An ATT&CK/CAPEC knowledge base names the technique and its mitigation; CVE exposure raises host risk.



### What-if defence simulator

Test a block or isolate action inside the model before using it.



### Cheap enough for CII

~170k parameters: 12 h of traffic forecast in seconds on a laptop CPU, fully offline.

### Built: flow + packet pipeline

PCAP (Scapy) + CIC / CTU-13 / UNSW loaders

### Built: world model + forecaster

K-step rollouts, Shapley, what-if

### Built: offline demo app

Streamlit + CLI + signed receipts

### Next: real-data training

CIC-IDS2018, CTU-13, DARPA 2000; CII pilot



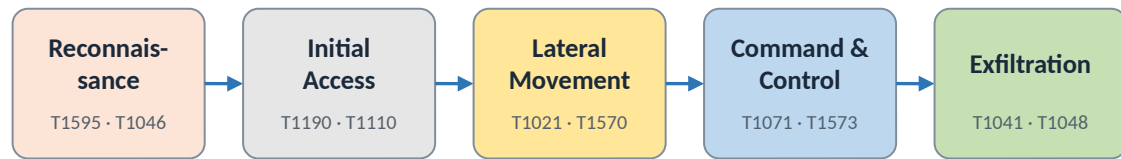
# TECHNICAL APPROACH

## WORLD MODEL ARCHITECTURE

NetSentinel Foresight learns **how network state changes**. An encoder turns each time window into a state, a dynamics model learns  $P(S_{t+1} | S_t)$ , and a rollout engine simulates K steps ahead.

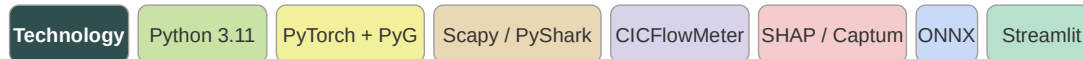
- **State  $S_t$  (every 60 s window):** a graph with hosts as nodes and flows as edges. Flow-level features: TCP flag counts (SYN/ACK/FIN/RST/PSH/URG), bytes, packets, duration, IAT mean/var/max, bidirectional ratio. Packet-level features: TTL mean and variance, TCP window size, fragment flags, payload-size histogram, sequential vs random port order, retransmissions.
- **Encoder:** edge-weighted GraphSAGE over the host graph + an MLP on the state vector  $\rightarrow$  one latent vector per window, plus one per host.
- **Dynamics (the world model):** recurrent state-space model (GRU memory + stochastic latent) with a causal Temporal Transformer over the last 30 windows. It learns to predict the next state's features (self-supervised) and the next ATT&CK stage from the datasets' attack timelines (supervised dynamics learning).
- **Forward simulation:** sample 64 latent paths  $K = 1 \dots 10$  windows ahead  $\rightarrow$   $P(\text{infiltration within } K)$ , predicted stage and next target host.
- **Explanations:** attention over time and neighbours + SHAP over feature groups  $\rightarrow$  e.g. "top drivers: SYN-only ratio  $\uparrow$ , dst-port entropy  $\uparrow$ , TTL variance  $\uparrow$ ".

## FORECAST ATTACK STAGES (MITRE ATT&CK)

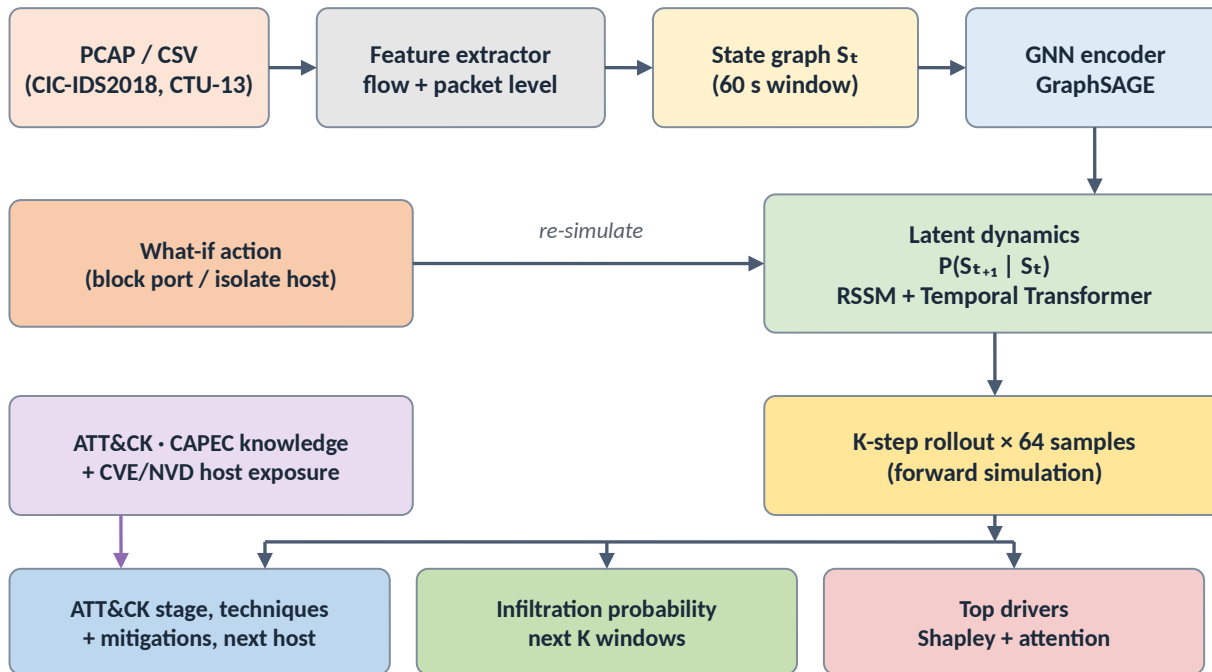


Dataset labels (e.g. CIC-IDS2018 "Infiltration", "Brute Force", "Bot"; DARPA 2000 LLDOS phases 1-5) are mapped to these stages. The benign  $\rightarrow$  recon  $\rightarrow$  access transitions are the ones the model must learn to see coming.

## NETSENTINEL TECH STACK



## WORLD-MODEL PIPELINE



Every forecast is signed into our Merkle ledger, so it can be checked later against what actually happened.

**Supervised dynamics learning:** ground-truth transitions come from each dataset's attack timeline. The loss weights transitions into attack stages, so rare stages are learned. **Output per window:**  $P(\text{infiltration in the next } K \text{ windows})$ , predicted stage, top contributing flags/ports/flow stats, and the next target host. **Not a static classifier:** we show it by shuffling time. If accuracy does not drop, it has not learned dynamics.



# FEASIBILITY & VIABILITY

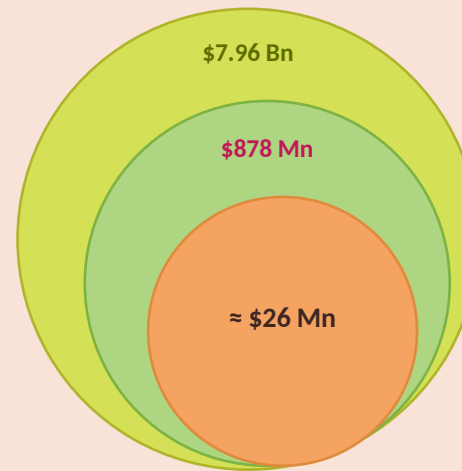
## POTENTIAL CHALLENGES AND RISKS:

**Attack windows are rare:** attack time is a small fraction of each capture, and stages like exfiltration are rarer still.

**Order may carry no signal:** our earlier project found shuffling time changed nothing. Here we test it on every result: shuffling history drops F1 from 0.865 to 0.347.

**Dataset shift:** a model trained on a lab network may not carry over to a real enterprise or CII network.

### Market Opportunity Funnel



**TAM \$7.96 Bn**

Global Intrusion Detection & Prevention market, 2026  
(Precedence Research)

**SAM up to \$878 Mn**

India OT security market, 2026  
(MarketsandMarkets)

**SOM ≈ \$26 Mn**

We aim for 3% of the available market: SOC's that want early warning, not only alerts.

## STRATEGIES TO OVERCOME THESE CHALLENGES:

- **Weight the rare stages:** focal loss, oversampling of attack transitions, training on several datasets (CIC-IDS2018, CTU-13, UNSW-NB15, CICIOT2023).
- **Prove the time effect:** every result is shown next to logistic regression on the same features and next to our model with time shuffled. If there is no gap, we say so.
- **No-leak, unseen-attack tests:** split by time, never by random row. Hold out whole attack families and whole datasets. Fine-tune per site on ~10 days of its own benign traffic.

| Component                               | Trained / tested on              | What we will report                                   |
|-----------------------------------------|----------------------------------|-------------------------------------------------------|
| Feature pipeline · Scapy + CICFlowMeter | CIC-IDS2018 CSV + PCAP, CTU-13   | Timestamped, normalised flow + packet matrix; flows/s |
| World model · RSSM + Transformer + GNN  | CIC-IDS2018, CTU-13, UNSW-NB15   | Next-state error; stage F1 at $t+1 \dots t+K$         |
| Forecast engine · K-step rollout        | DARPA 2000 LLDOS, LANL red team  | Lead time before compromise; AUC@K; Brier score       |
| Baseline · Logistic regression          | Same features, same time split   | F1, precision, recall, FPR, side by side              |
| Generalisation                          | Leave-one-attack-out; CICIOT2023 | Recall on attack types never seen in training         |
| Explainer · SHAP + attention            | All of the above                 | Deletion test: remove top-5 drivers → risk must fall  |

## PROOF IT RUNS TODAY:

**Runs on a laptop CPU:** 12 h of traffic (67,650 flows) forecast in ~2 s: 720 windows x 64 simulated futures.

**PCAP in, offline:** a 441k-packet PCAP parsed into flow + packet features in 2.7 s. No GPU, no cloud.

**Beats the baseline:** F1 0.865 vs 0.483 (logistic regression) at the same 3.1% false-positive rate.\*

**Tested:** 28 automated tests: loaders, PCAP round-trip, Shapley efficiency, tamper detection.

*\*Current scores are on held-out simulated scenarios (bundled simulator). This table is the real-data plan.*



# IMPACT & BENEFITS



SMART INDIA  
HACKATHON  
2026

## NETSENTINEL VS. EXISTING TOOLS

| Capability                                     | Forecasts next stage             | Learns time dynamics   | ATT&CK stage mapping | Explains each output  | Works offline       |
|------------------------------------------------|----------------------------------|------------------------|----------------------|-----------------------|---------------------|
| NetSentinel Foresight                          | Yes: K-step probability          | Yes: world model       | Yes, per forecast    | Yes: SHAP + attention | Yes                 |
| Snort / Suricata                               | No                               | No: rules              | Partial (rule tags)  | Rule name only        | Yes                 |
| ML flow classifiers (RF / XGBoost IDS)         | No                               | No: one flow at a time | No                   | Sometimes             | Yes                 |
| Commercial NDR / SIEM (e.g. Darktrace, Splunk) | Risk scores, not stage forecasts | Varies                 | Yes                  | Partial               | Varies: often cloud |

## IMPACTS

**The blind spot:** SOC's protecting India's power, banking, telecom and government networks see an attack only after a stage completes, and CERT-In's 6-hour reporting clock starts after the damage is done.

**What changes:** a forecast minutes to hours ahead, naming the next stage and the host likely hit next.

**For operators:** an early warning while the attacker is still scanning.

**For analysts:** one forecast carrying the whole chain and its drivers, not scattered alerts.

**For auditors:** every forecast is signed and later checked against reality, so accuracy itself is auditable.

**For the nation:** runs inside the premises of NCIIPC-notified CII. No traffic data leaves the site.

**Beyond this:** hospitals, rail signalling, smart grids and IoT (tested on CIIoT2023).

## USE CASES



### Power & Energy Grid (CII)

Forecasts recon → lateral movement towards SCADA hosts before control is reached.



### Government SOC & CERT-In

ATT&CK-mapped forecasts and signed evidence fit the 6-hour reporting rule.



### Banking & Finance

Predicts brute-force → access → exfiltration and names the server at risk.



### Enterprises, MSMEs & Colleges

CPU-only, offline, one SPAN port: early warning they can afford.



### Security Benefits

**Act before compromise:** block, isolate or watch while the attacker is still in recon.

**Knows the next target:** response is focused on the host most likely hit next.

**Catches the unseen:** judges stage behaviour, not signatures.



### Economic Benefits

**Cheaper to stop early:** containing at recon costs far less than cleaning up a breach.

**Low hardware cost:** one CPU box per segment, no GPU, no licence fees.

**Less alert fatigue:** one forecast per campaign, not dozens of alerts.



### Technical Benefits

**Offline & open source:** code, weights and training configs included, reproducible.

**Two levels of traffic:** flow + packet features fused on a graph.

**What-if simulation:** try a defensive action in the model first.



# RESEARCH & REFERENCES

## CITATIONS

### World Models

**Ha & Schmidhuber, 2018**

Learn an internal model, then roll it forward.

### Latent Dynamics (RSSM)

**Hafner et al., DreamerV3, 2023**

The recurrent state-space model we adapt to network state.

### Graph NIDS

**Lo et al., E-GraphSAGE, IEEE NOMS 2022**

Flows as graph edges; basis of our encoder.

### Attack Forecasting

**Husák et al., IEEE COMST 2019**

Survey of attack projection, prediction & forecasting.

### Explainability

**Lundberg & Lee, NeurIPS 2017 (SHAP)**

Feature attribution for every forecast.

## LINKS (Ctrl + Click on the blue words)



[Demo Video - Prototype](#)



[GitHub Repository](#)



[Documentation](#)



[Presentation](#)

## NetSentinel Foresight · Standards & Knowledge Alignment

### MITRE ATT&CK

Stage labels + technique IDs for every forecast (T1595, T1190, T1021, T1071, T1041...).

### CAPEC

Attack patterns linked to every ATT&CK technique the evidence points to.

### CVE / NVD

CVSS exposure of each host's open services raises its chance of being next.

### NCIIPC · CII

Design fits NCIIPC guidance for protected systems: on-prem, passive, no data leaves.

### CERT-In

ATT&CK-mapped, signed forecasts support the 6-hour incident reporting rule.

### RFC 6962

Merkle transparency log (as used for web certificates) for tamper-evident forecast receipts.

## DATASET IMPLEMENTATION

1. CIC-IDS2017/2018: flows + PCAP, day-wise attack timelines

2. CTU-13: 13 botnet scenarios, labelled NetFlow + PCAP

3. UNSW-NB15: 9 attack families, PCAP + flows

4. CICIOT2023: 33 IoT attacks (unseen-attack tests)

5. DARPA 2000 LLDOS 1.0/2.0: 5-phase multi-stage attack

6. LANL auth + red team: real lateral movement, 58 days

7. MITRE ATT&CK · CAPEC · CVE/NVD: knowledge base

8. Own capture, IPv4 + IPv6: benign baseline (already collected)

## MODEL SELECTION — HEAD-TO-HEAD

| Component                               | Compared against                                      | Pass condition                                                                 |
|-----------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>World model · RSSM + Transformer</b> | Logistic regression; same model with history shuffled | <b>Done*</b> : F1 0.865 vs 0.483 at equal FPR; shuffled history drops to 0.347 |
| <b>Graph encoder · GraphSAGE</b>        | Flat feature vector                                   | <b>Better next-target</b> accuracy from who-talks-to-whom                      |
| <b>Rollout · 64 samples</b>             | Single one-step prediction                            | <b>Calibrated</b> probabilities (Brier / ECE)                                  |
| <b>Explainer</b>                        | Attention only · SHAP · Integrated Gradients          | <b>Faithful</b> : removing top drivers drops the risk                          |
| <b>ATT&amp;CK / CAPEC knowledge</b>     | Stage label only                                      | <b>Actionable</b> : each alarm names technique, CAPEC pattern and mitigation   |

\*Measured on held-out simulated scenarios (results/benchmark.md). Other rows: planned, same features and time split, published win or lose.